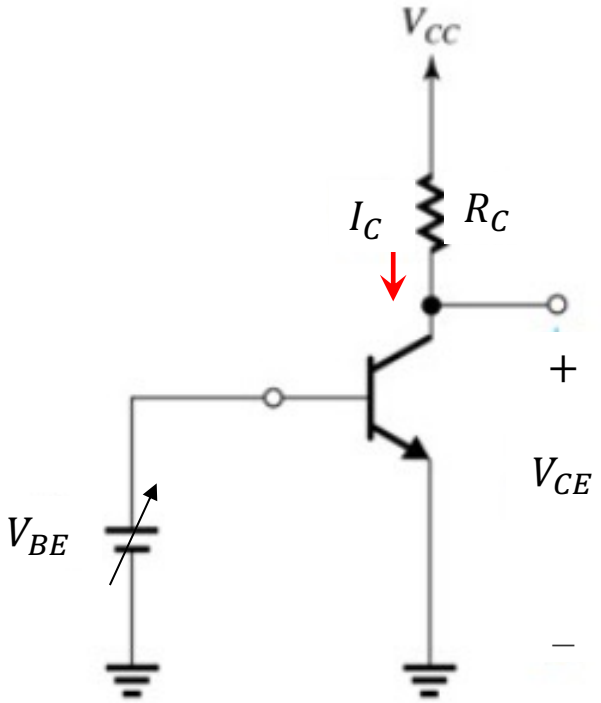
Bipolar Junction Transistors

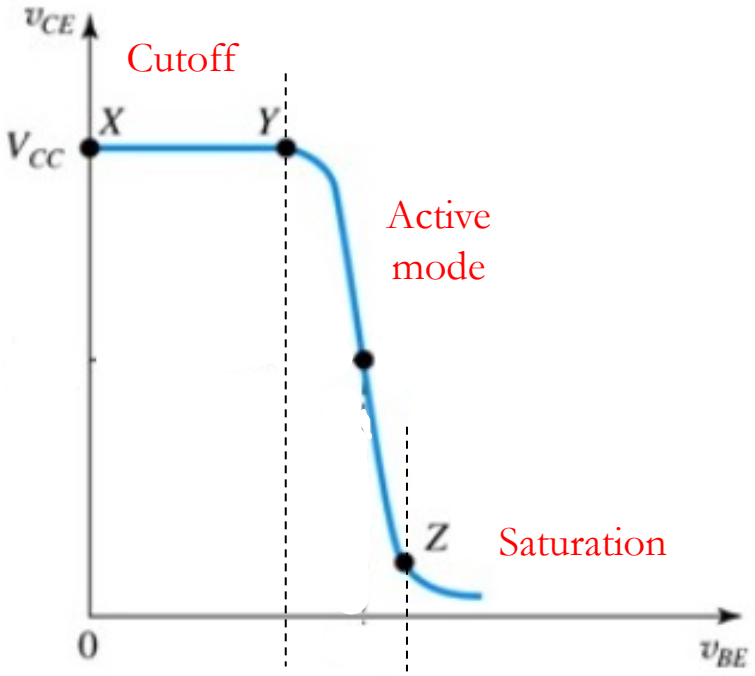
- Small Signal Analysis.
- The BJT as an Amplifier.

The BJT Transfer Function

The transfer function indicated the region of operation of each mode. It also help identify the possible applications.

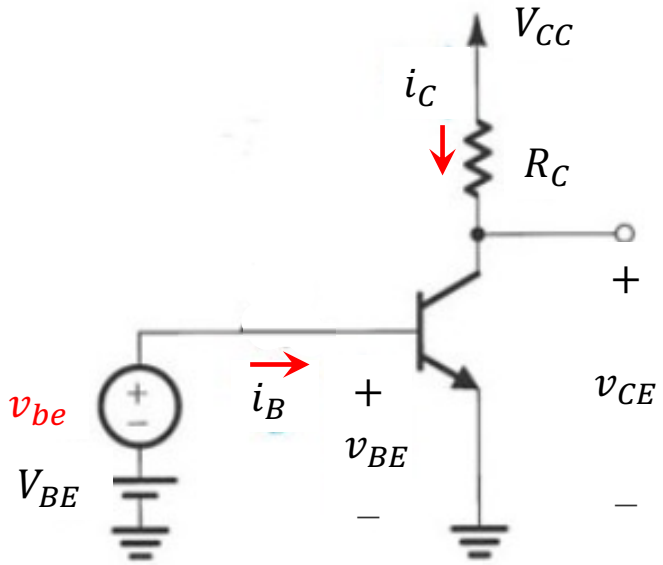
$$V_{CE} = V_{CC} - I_C R_C$$





Small Signal Analysis of BJTs

Let's consider the circuit below, where both DC and ac signals (large signal) are applied. We try to write the circuit variables as two components: DC and ac.



$$v_{BE} = V_{BE} + v_{be}$$

$$i_C = I_s\, e^{v_{BE}/V_T} = I_s\, e^{V_{BE}/V_T} e^{v_{be}/V_T}$$

$$i_C = I_C\, e^{v_{be}/V_T}$$

Small-signal approximation: $v_{be} \ll V_T$

$$i_C = I_C\,(1 + \frac{v_{be}}{V_T})$$

$$i_C = I_C + \frac{I_C}{V_T} v_{be}$$

DC component

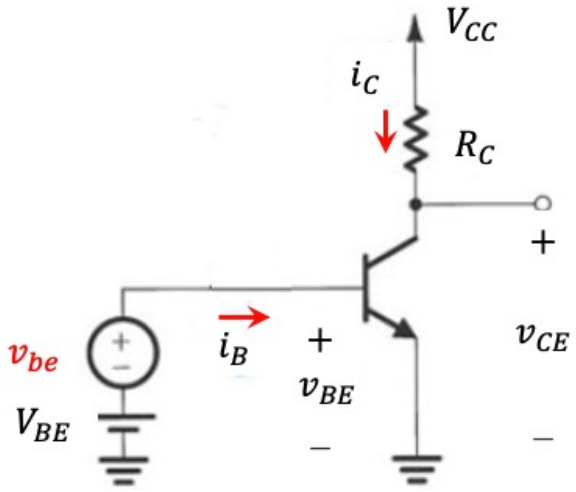
ac component

$$i_c = \frac{I_C}{V_T} v_{be} = g_m v_{be}$$

g_m Transconductance (A/V)

Small Signal Analysis

Let's consider the circuit below, where both DC and ac signals (large signal) are applied. We try to write the circuit variables as two components: DC and ac.



$$i_C = I_C + \frac{I_C}{V_T} v_{be} \qquad i_c = g_m v_{be}$$

$$i_B = \frac{I_C}{\beta} + \frac{I_C}{\beta V_T} v_{be} \qquad i_b = \frac{g_m}{\beta} v_{be}$$

$$i_E = \frac{I_C}{\alpha} + \frac{I_C}{\alpha V_T} v_{be} \qquad i_e = \frac{g_m}{\alpha} v_{be}$$

$g_m = \frac{I_C}{V_T}$ (A/V)
Transconductance

$r_\pi = \frac{\beta}{g_m}$ (Ω)
Base resistance

$r_e = \frac{\alpha}{g_m}$ (Ω)
Emitter resistance

$r_o = \frac{V_A}{I_C}$ (Ω)
Output resistance

Note that r_π and r_e both represent the base-emitter circuit; one looking from the base side (r_π) and the other looking from the emitter side (r_e). Also,

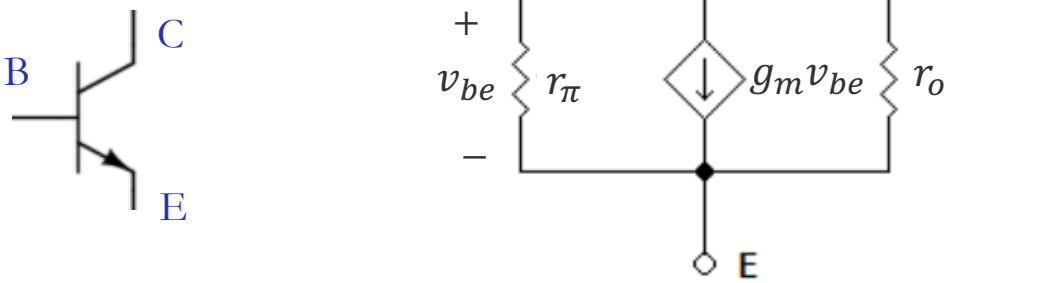
$$r_\pi = (\beta + 1)\, r_e$$

Based on these equations, we can develop a small-signal equivalent circuit of the BJT transistor.

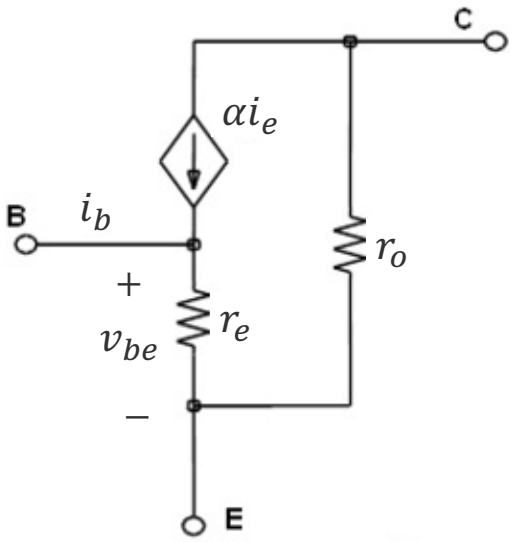
BJT Small-signal Equivalent Circuit

Based on these equations, we can develop a small-signal equivalent circuit of the BJT transistor.

The pi-model: involves the base resistance



The T-model: involves the emitter resistance



Small-signal parameters of the BJT

$$g_m = \frac{I_C}{V_T} \text{ (A/V)}$$

Transconductance

$$r_\pi = \frac{\beta}{g_m} \text{ } (\Omega)$$

Base resistance

$$r_e = \frac{\alpha}{g_m} \text{ } (\Omega)$$

Emitter resistance

$$r_o = \frac{V_A}{I_C} \text{ } (\Omega)$$

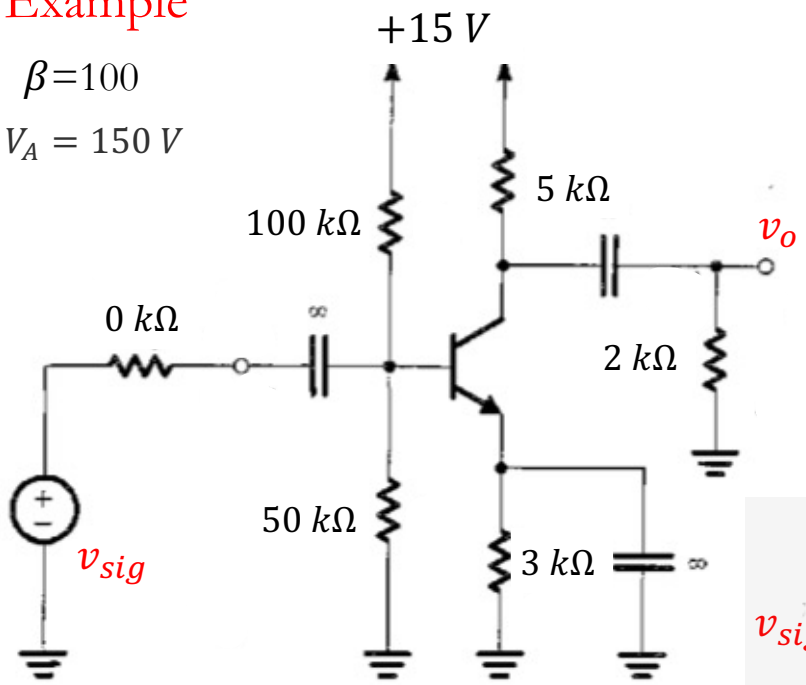
Output resistance

Small Signal Analysis

Example

β=100

$V_A = 150\ V$



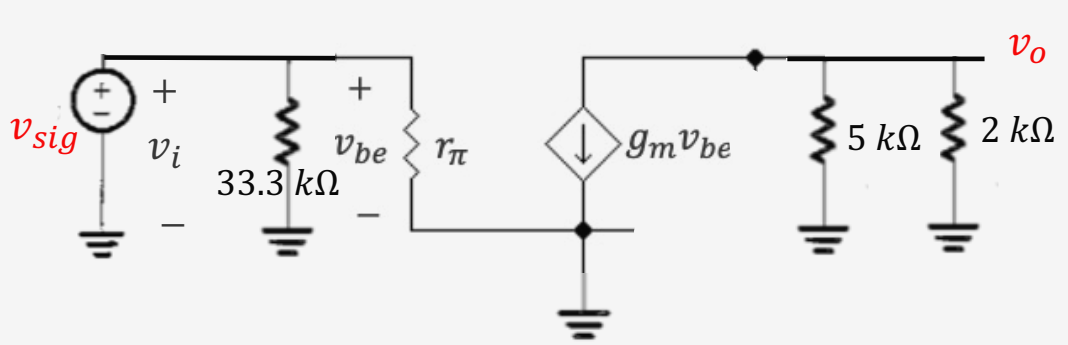
DC Analysis: Refer to week 11 examples

$$I_C = 1.28\ mA$$

$$g_m = \frac{I_C}{V_T} = 50\ \text{mA/V} \qquad r_\pi = \frac{\beta}{g_m} = 2\ k\Omega$$

$$r_e = \frac{\alpha}{g_m} = 19.8\ \Omega \qquad r_o = \frac{V_A}{I_C} = 117\ k\Omega$$

AC Analysis: Use pi-model, ignore r_o



$$v_o = -g_m v_{be}(5 \parallel 2)k$$

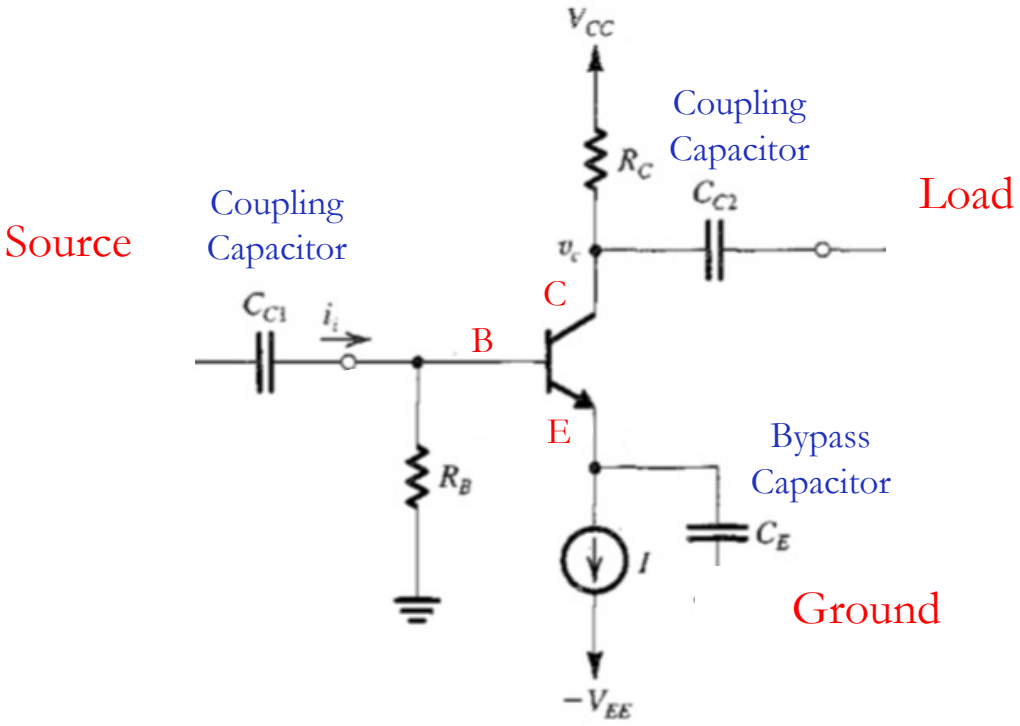
$$v_i = v_{be}$$

$$A_v = \frac{v_o}{v_i} = -g_m(5 \parallel 2)k = -71.4\ (V/V)$$

$$R_i = (33.3 \parallel r_\pi)k = 1.88\ k\Omega$$

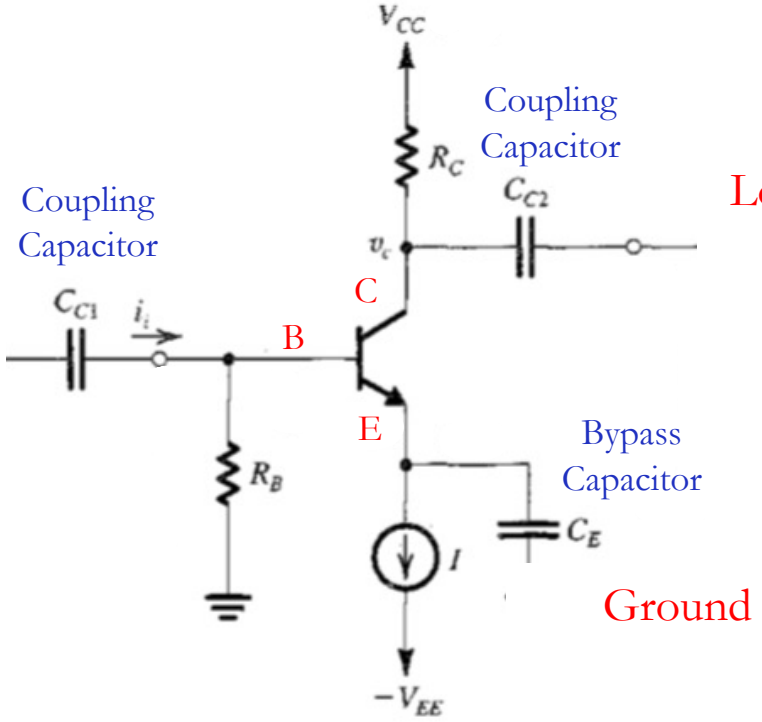
$$R_o = 5\ k\Omega$$

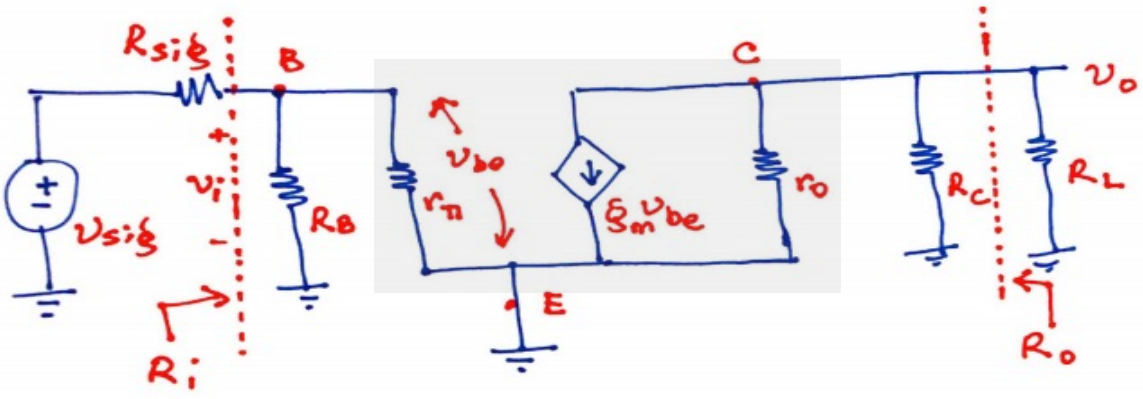
Basic BJT Amplifier Configurations



1) The common-emitter (CE) amplifier
2) The common-base (CB) amplifier
3) The common-collector (CC) amplifier

The Common-emitter (CE) Amplifier





The Common-emitter (CE) Amplifier

The small-signal analysis goes as follows.

$$v_i = \frac{R_i}{R_i + R_{sig}} v_{sig}$$

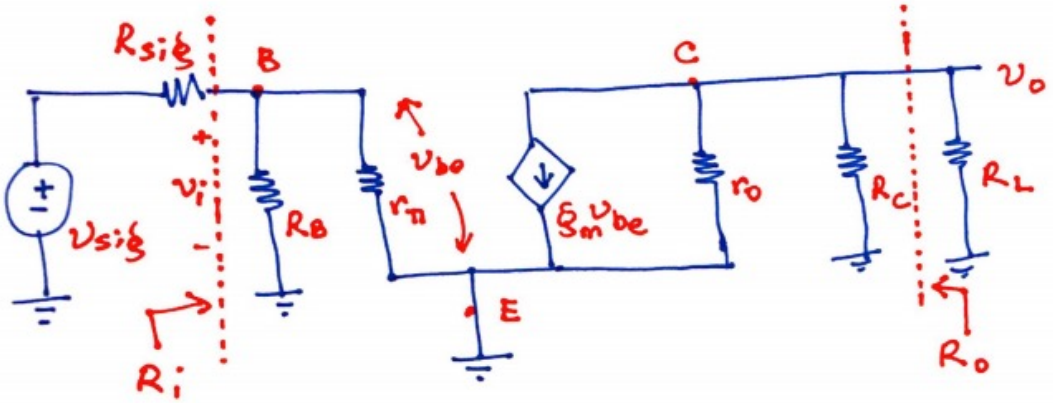
$$v_{be} = v_i$$

$$v_o = -g_m v_{be}(r_o \,//\, R_C \,//\, R_L)$$

$$\frac{v_o}{v_i} = -g_m\,(r_o \,//\, R_C \,//\, R_L)$$

$$A_m = \frac{v_o}{v_{sig}} = -g_m \frac{R_i}{R_i + R_{sig}}(r_o \,//\, R_C \,//\, R_L)$$

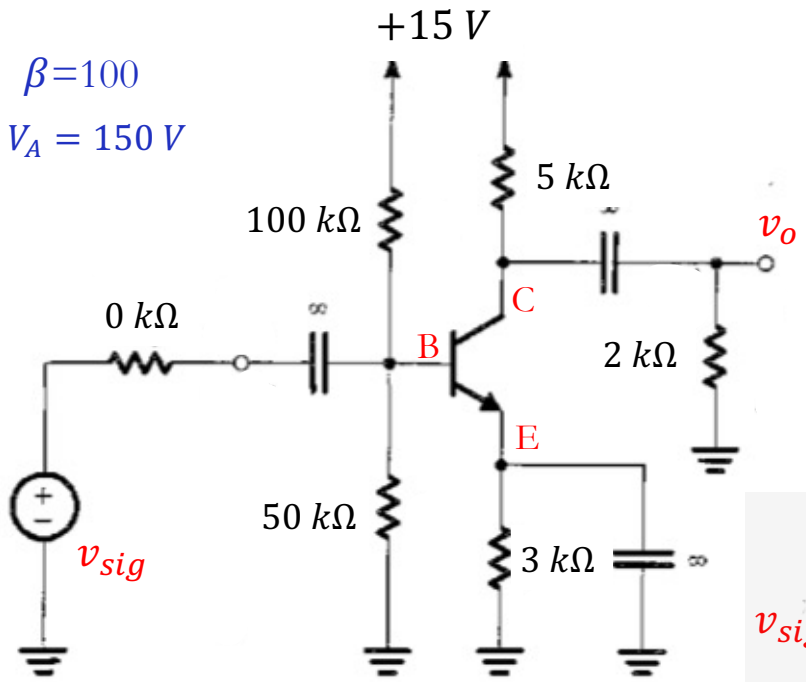
β-dependent



$$R_i = (r_\pi \,//\, R_B) \quad \text{Low}$$

$$R_o = (r_o \,//\, R_C) \approx R_C$$

Example



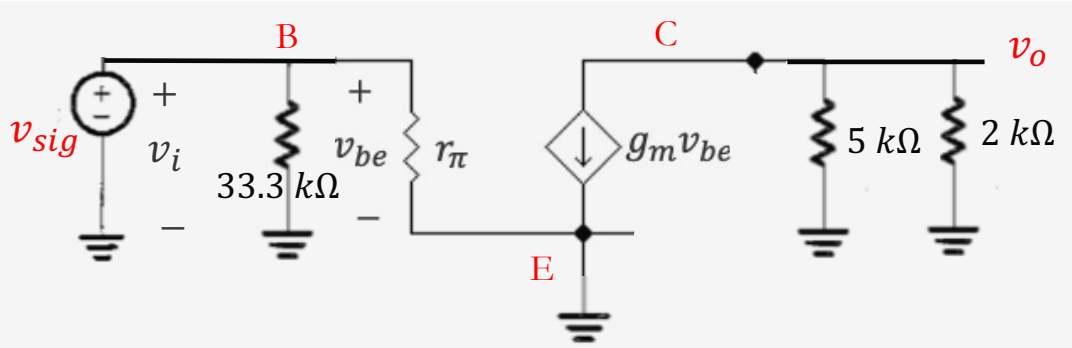
DC Analysis

$$I_C = 1.28\,mA$$

$$g_m = \frac{I_C}{V_T} = 50\ \text{mA/V} \qquad r_\pi = \frac{\beta}{g_m} = 2\ k\Omega$$

$$r_e = \frac{\alpha}{g_m} = 19.8\ \Omega \qquad r_o = \frac{V_A}{I_C} = 117\ k\Omega$$

AC Analysis: Use pi-model, ignore r_o



$$v_o = -g_m v_{be}(5 \parallel 2)k$$

$$v_i = v_{be}$$

$$A_v = \frac{v_o}{v_i} = -g_m(5 \parallel 2)k = -71.4\ (V/V)$$

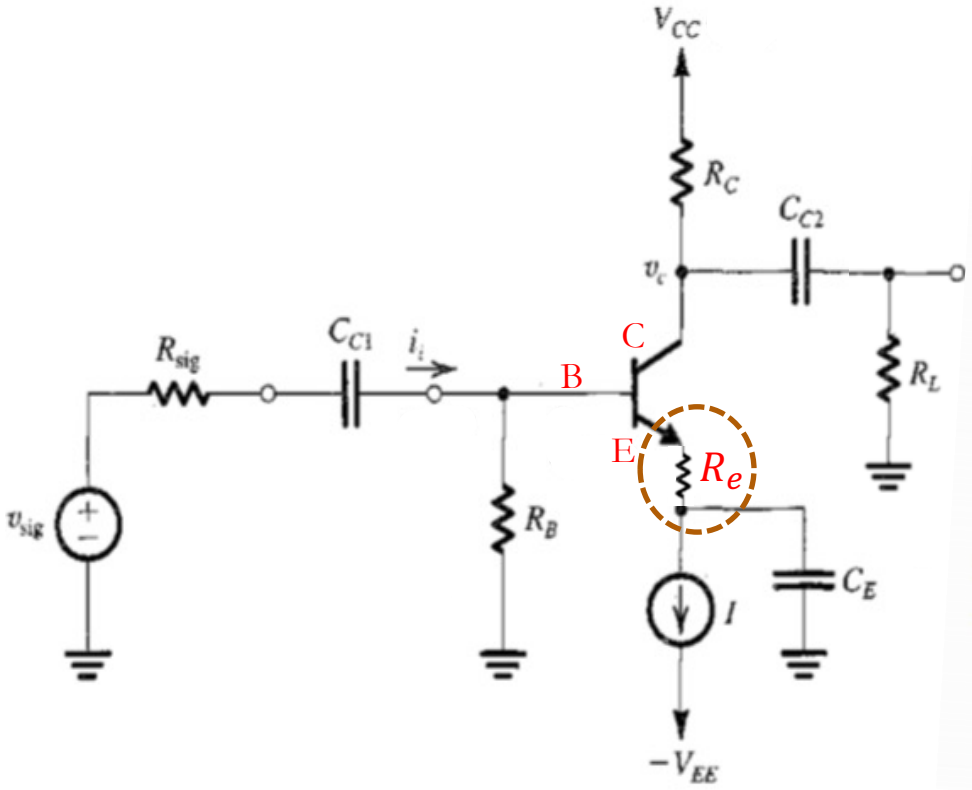
$$R_i = (33.3\ k \parallel r_\pi) = 1.88\ k\Omega$$

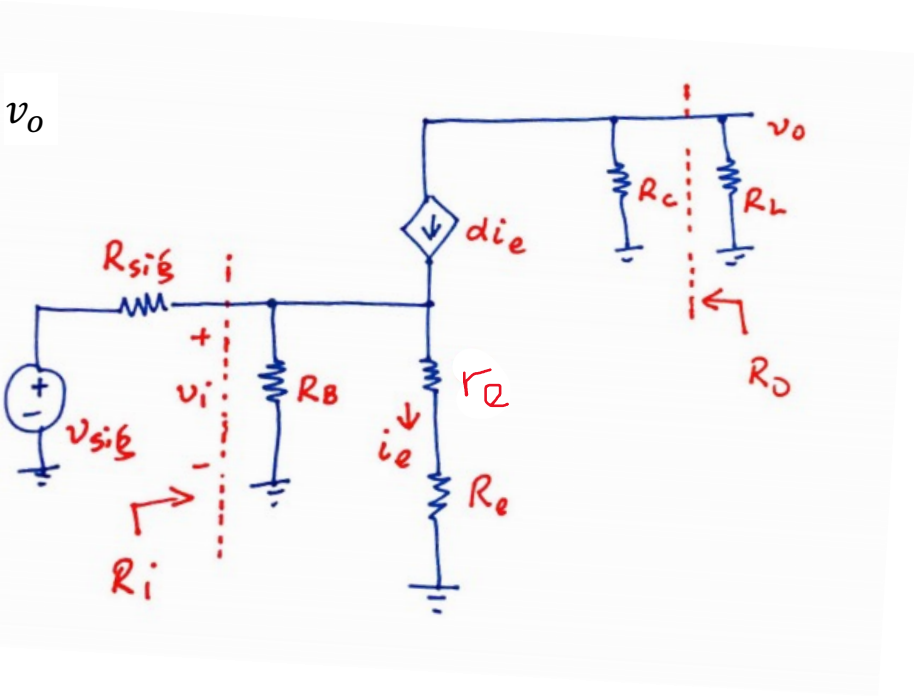
$$R_o = 5\ k\Omega$$

The CE Amplifier With R_e

A resistor is inserted in the emitter circuit.

- Reduces β dependence.
- Increases input resistance.
- Improves Bandwidth.
- Reduces gain.





The CE Amplifier With R_e

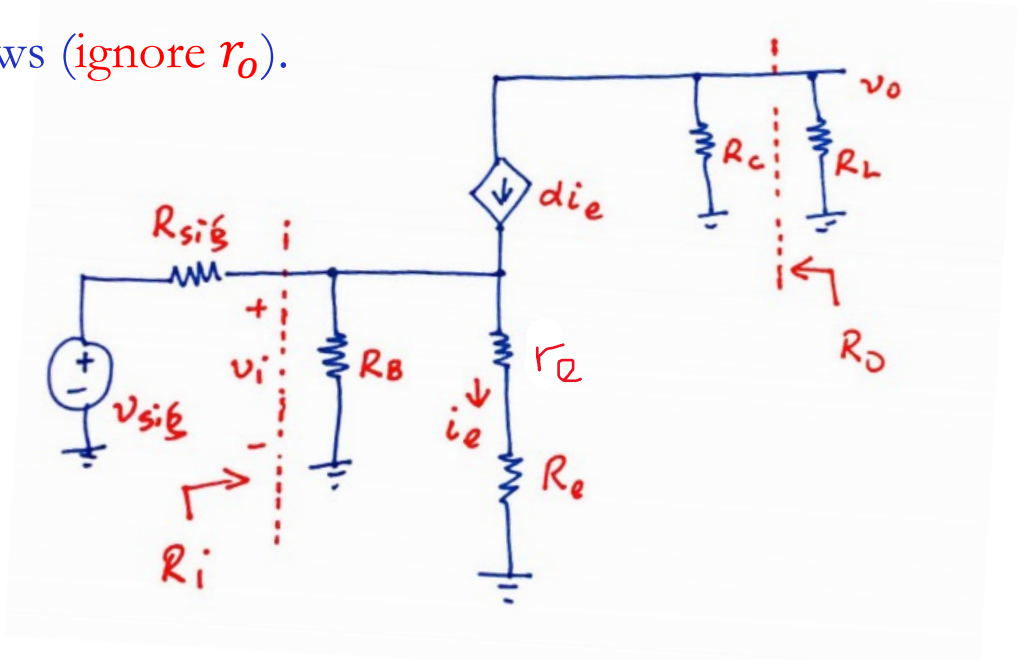
The small-signal analysis goes as follows (ignore r_o).

$$v_o = -\alpha\, i_e\, (R_C \,//\, R_L)$$

$$v_i = i_e\, (r_e + R_e)$$

$$v_i = \frac{R_i}{R_i + R_{sig}} v_{sig}$$

$$\frac{v_o}{v_i} = \frac{-\alpha\, (R_C \,//\, R_L)}{(r_e + R_e)}$$



Using $\alpha = g_m r_e$, we can write

$$\frac{v_o}{v_i} = \frac{-g_m (R_C \,//\, R_L)}{(1 + R_e / r_e)}$$

$$\frac{v_o}{v_i} \approx \frac{-g_m (R_C \,//\, R_L)}{(1 + g_m R_e)}$$

$$A_v = \frac{v_o}{v_{sig}} = -g_m \frac{R_i}{R_i + R_{sig}} \frac{(R_C \,//\, R_L)}{(1 + g_m R_e)}$$

Gain reduction factor

$$R_i = (r_\pi \,//\, R_B)(1 + g_m R_e)$$

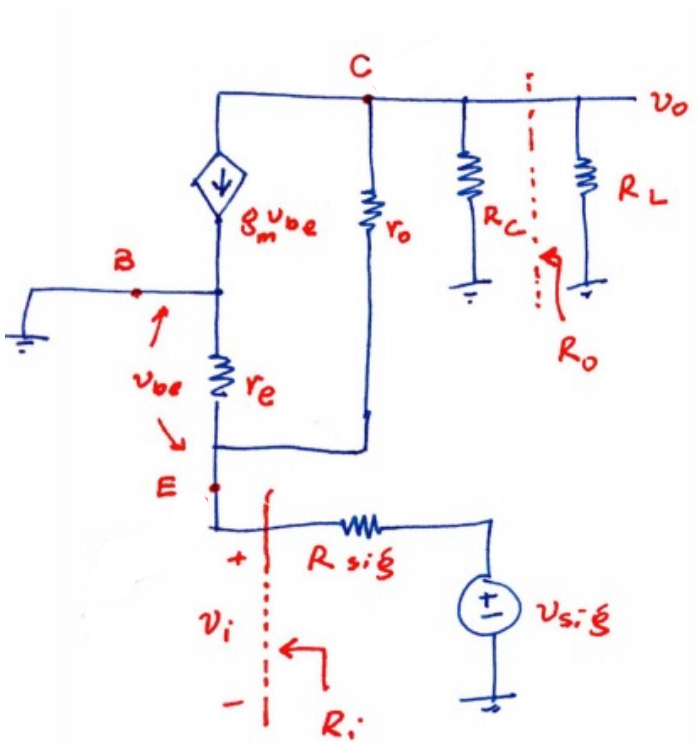
Input resistance increase factor

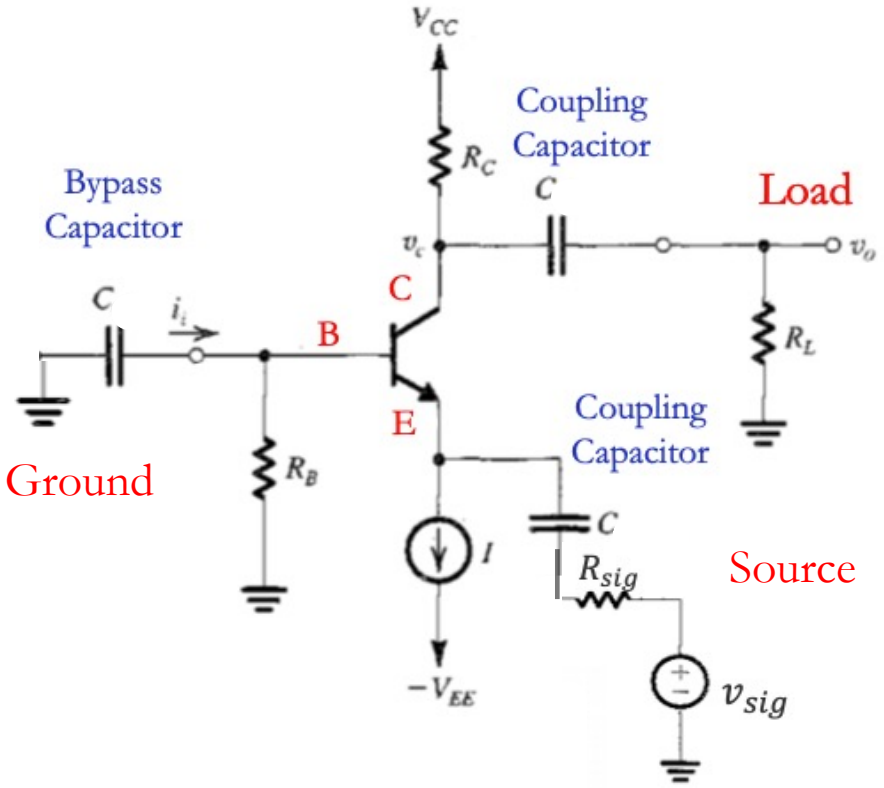
$$R_o \approx R_C$$

The Common-base Amplifier

Note the following:

- Because of the ground at the base, R_B will be shorted.
- The T-model equivalent circuit of the BJT will be used.





The Common-base Amplifier

The small-signal analysis goes as follows.

We can write KCL at 1 as,

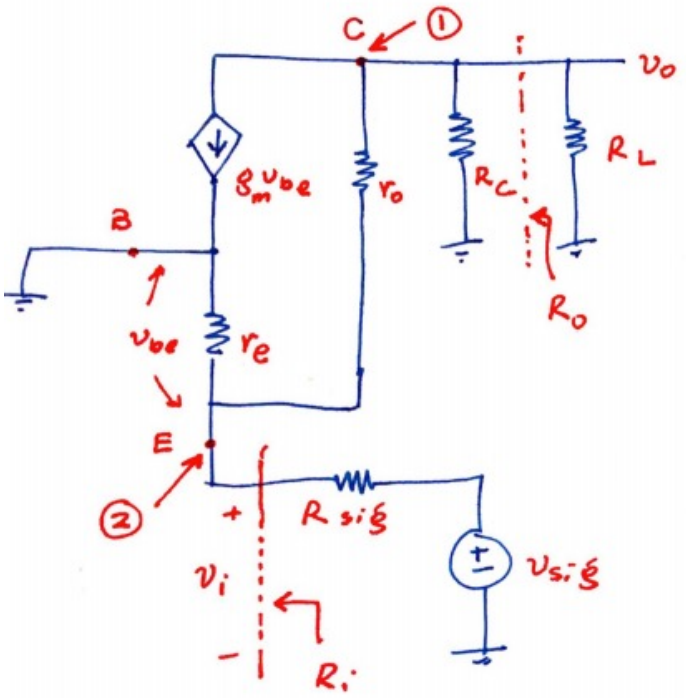
$$\frac{v_o - v_i}{r_o} + \frac{v_o}{R_C // R_L} + g_m v_{be} = 0$$

Note that $v_{be} = -v_i$

$$\Rightarrow \quad v_i\left(g_m + \frac{1}{r_o}\right) = v_o\left(\frac{1}{R_C // R_L} + \frac{1}{r_o}\right)$$

$$\frac{v_o}{v_i} = \frac{g_m + \frac{1}{r_o}}{\frac{1}{R_C // R_L} + \frac{1}{r_o}}$$

$$\frac{v_o}{v_i} \approx g_m(R_C // R_L)$$



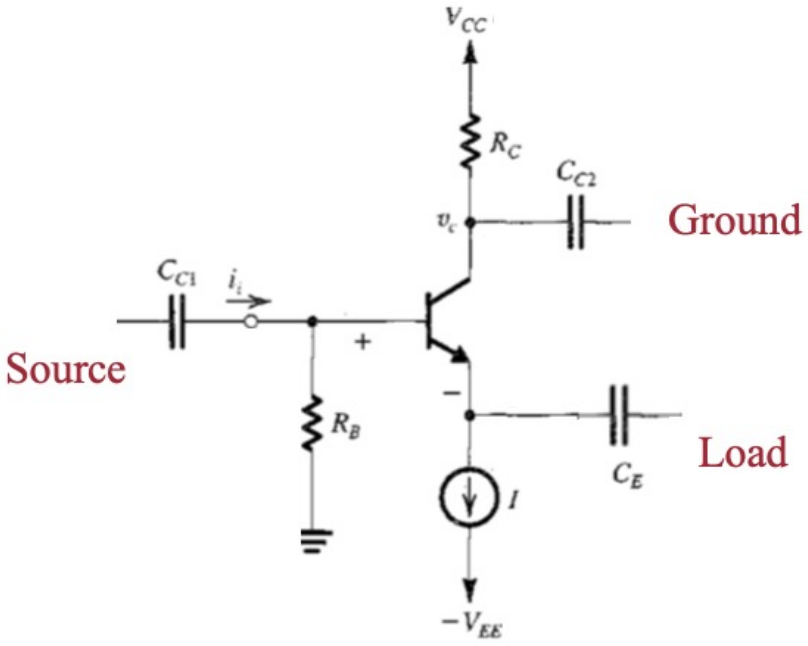
$$A_v = \frac{v_o}{v_{sig}} = g_m \frac{R_i}{R_i + R_{sig}}(R_C // R_L)$$

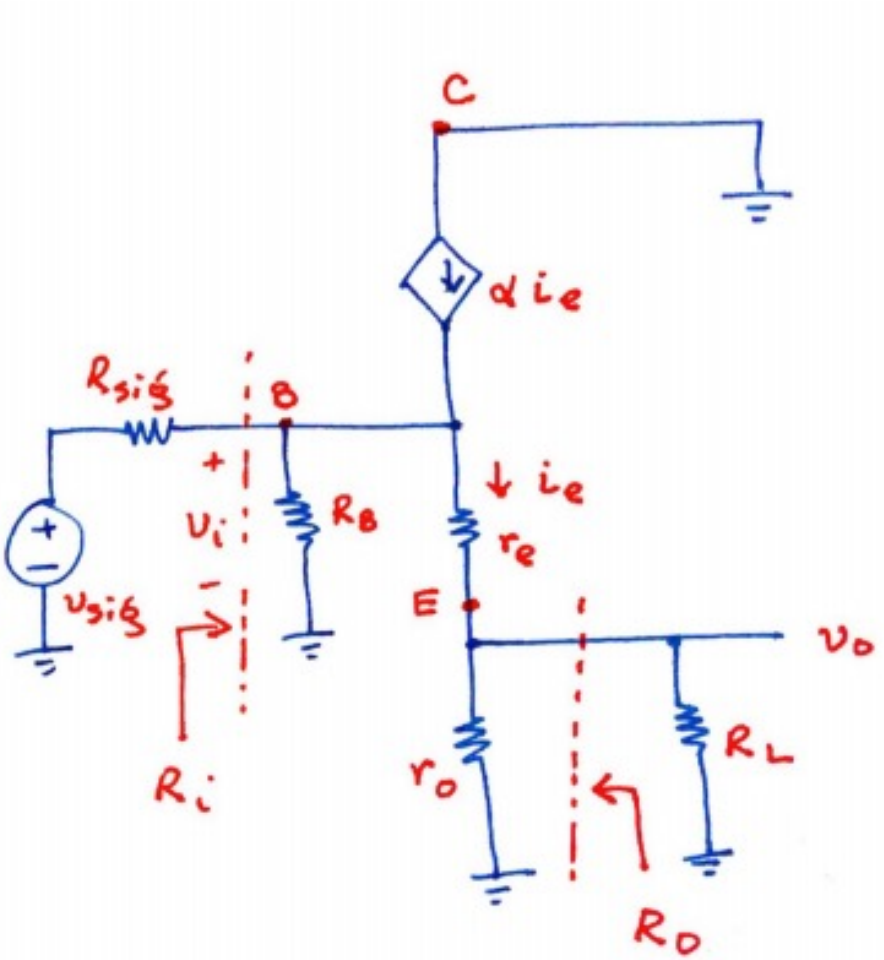
$$R_i = r_e$$

$$R_o \approx R_C$$

The Common-collector Amplifier

The small-signal equivalent circuit using the T-model becomes as below.





Note that r_o is pulled down because it is connected to ground, for clarity.

The Common-collector Amplifier

The small-signal analysis goes as follows.

We can write KVL from the input side as,

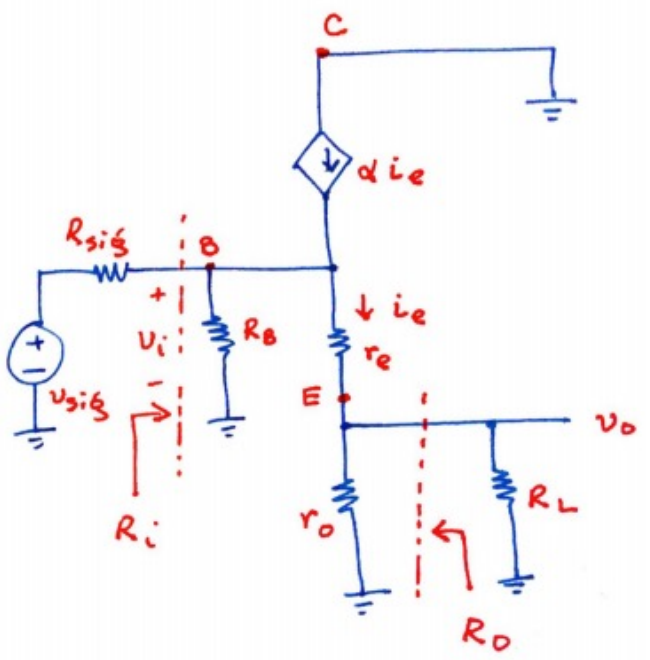
$$v_i = i_e r_e + v_o$$

$$v_i = \frac{v_o}{r_o \,//\, R_L} + v_o$$

$$v_i = v_o\left(1 + \frac{r_e}{r_o \,//\, R_L}\right)$$

$$\frac{v_o}{v_i} = \frac{r_o \,//\, R_L}{r_o \,//\, R_L + r_e}$$

$$A_v = \frac{v_o}{v_{sig}} = \frac{R_i}{R_i + R_{sig}} \frac{r_o \,//\, R_L}{r_o \,//\, R_L + r_e}$$



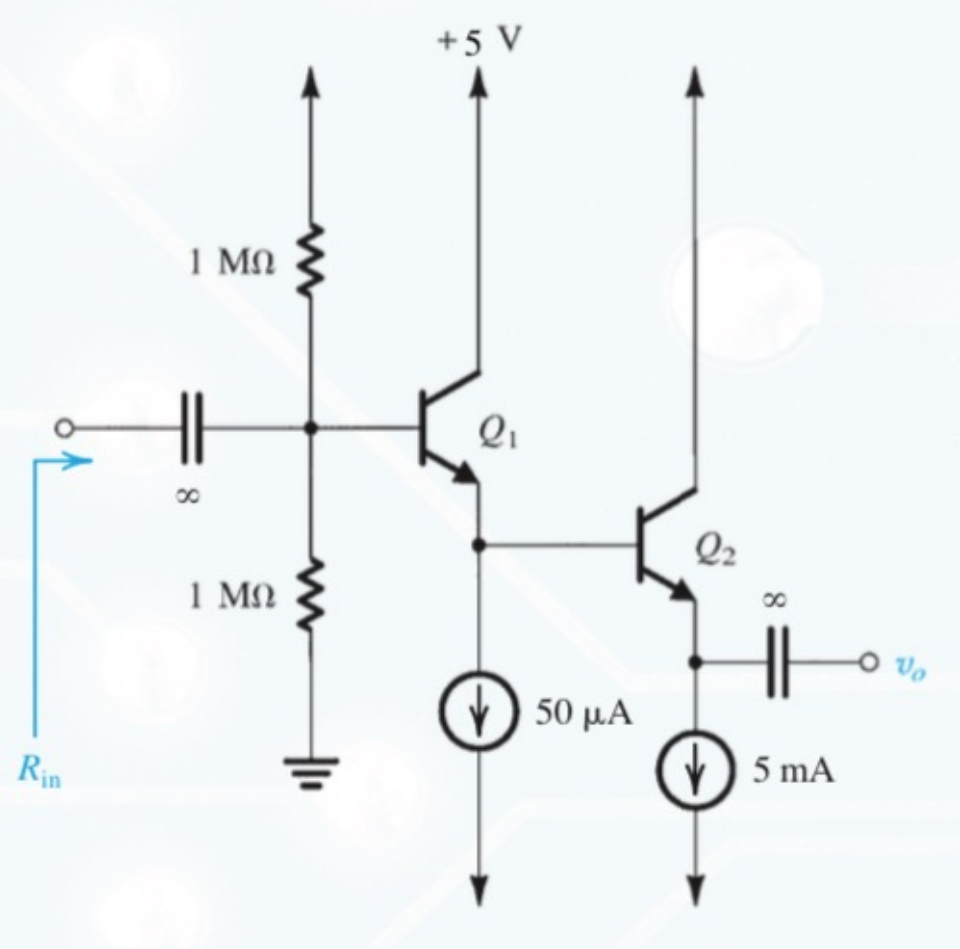
$$R_i = (\beta + 1)(r_o + r_e) \,//\, R_B \approx R_B$$

$$R_o = r_o \,//\, r_e \approx r_e$$

Examples

Analyze the amplifier circuit below.

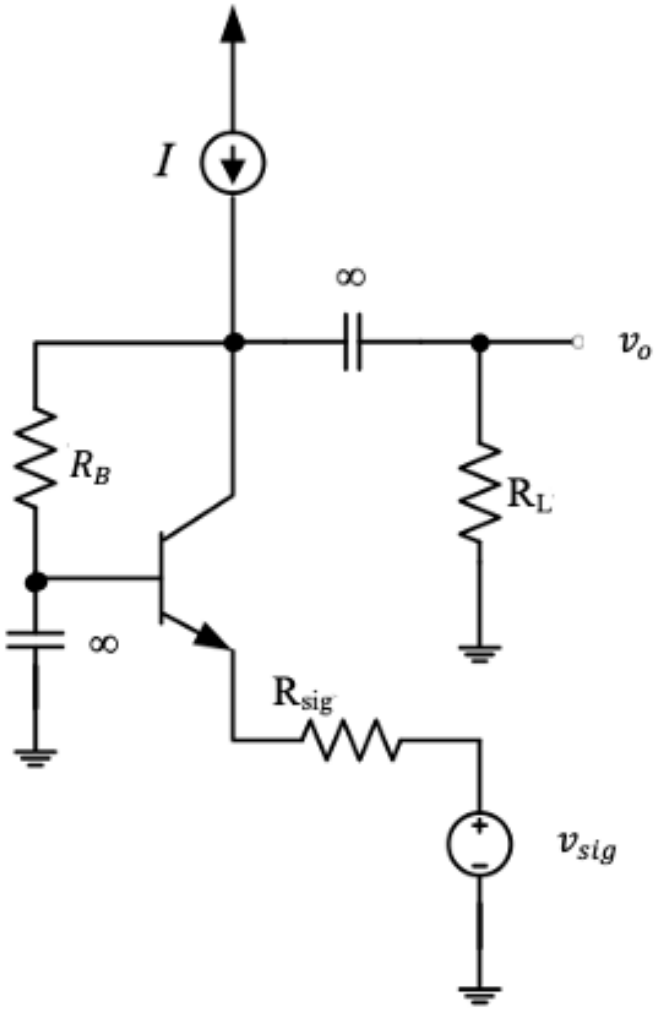
1) Identify the amplifier type.
2) Produce the small-signal equivalent circuit using the pi-model.
3) Find the amplifier voltage gain,
4) the current gain,
5) input resistance
6) output resistance.
7) Do the results confirm with the amplifier type? Explain. Ignore r_o.



Examples

Analyze the amplifier circuit below.

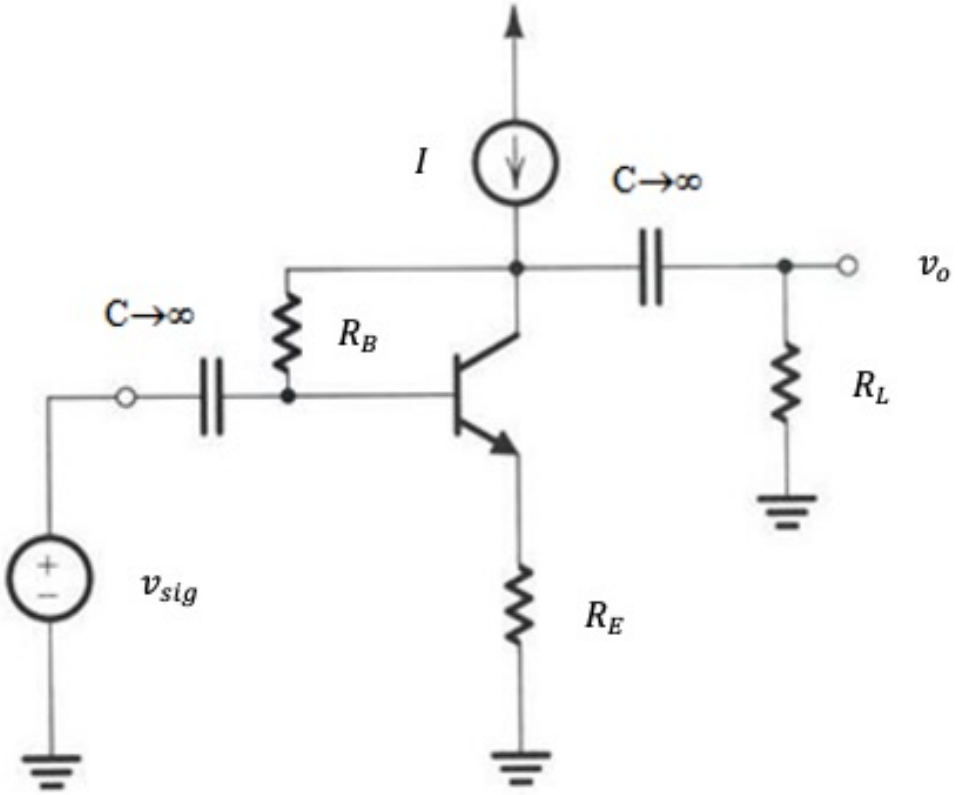
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Examples

Analyze the amplifier circuit below.

1) Identify the amplifier type.
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3) Find the amplifier voltage gain,
4) the current gain,
5) input resistance
6) output resistance.
7) Do the results confirm with the amplifier type? Explain. Ignore r_o.



Q & A

Send me an email if you have any questions.